BioHarvest-Sim: Analytical Assessment Report

Species: White-Tailed Deer | Generated: 2026-09-07 12:02

ISO/IEC/IEEE 12207 Compliance | High-Precision Linear Algebra Core

EXECUTIVE SUMMARY:
This report details the demographic stability and sustainable yield optimization for 'White-Tailed Deer' based on discrete-time Leslie matrix theory. The population exhibits a dominant eigenvalue λ_1 = 1.3967, corresponding to an intrinsic growth rate of r = 0.3341 per year. The net reproductive rate R_0 is 2.9885 with a mean generation time of 3.28 years. The theoretical maximum uniform sustainable harvest fraction is 28.40% per annum.

Demographic Parameter	Value
Age Classes (N)	6
Dominant Eigenvalue (λ_1)	1.39670
Intrinsic Growth Rate (r = ln λ_1)	0.33411 yr^-1
Net Reproductive Rate (R_0)	2.98854
Mean Generation Time (Tc)	3.28 years
Damping Ratio (λ_1 / $\lvert\lambda_2\rvert$)	1.8433
Perron-Frobenius Primitivity	Verified Primitive
Matrix Diagonalizable	Yes (L = P D P^-1)
Uniform Sustainable Harvest (h*)	28.40% per step

Leslie Projection Matrix (L):

```
[[0.15 0.85 1.45 1.6  1.5  0.9 ]
 [0.65 0.   0.   0.   0.   0.  ]
 [0.   0.82 0.   0.   0.   0.  ]
 [0.   0.   0.85 0.   0.   0.  ]
 [0.   0.   0.   0.8  0.   0.  ]
 [0.   0.   0.   0.   0.75 0.  ]]
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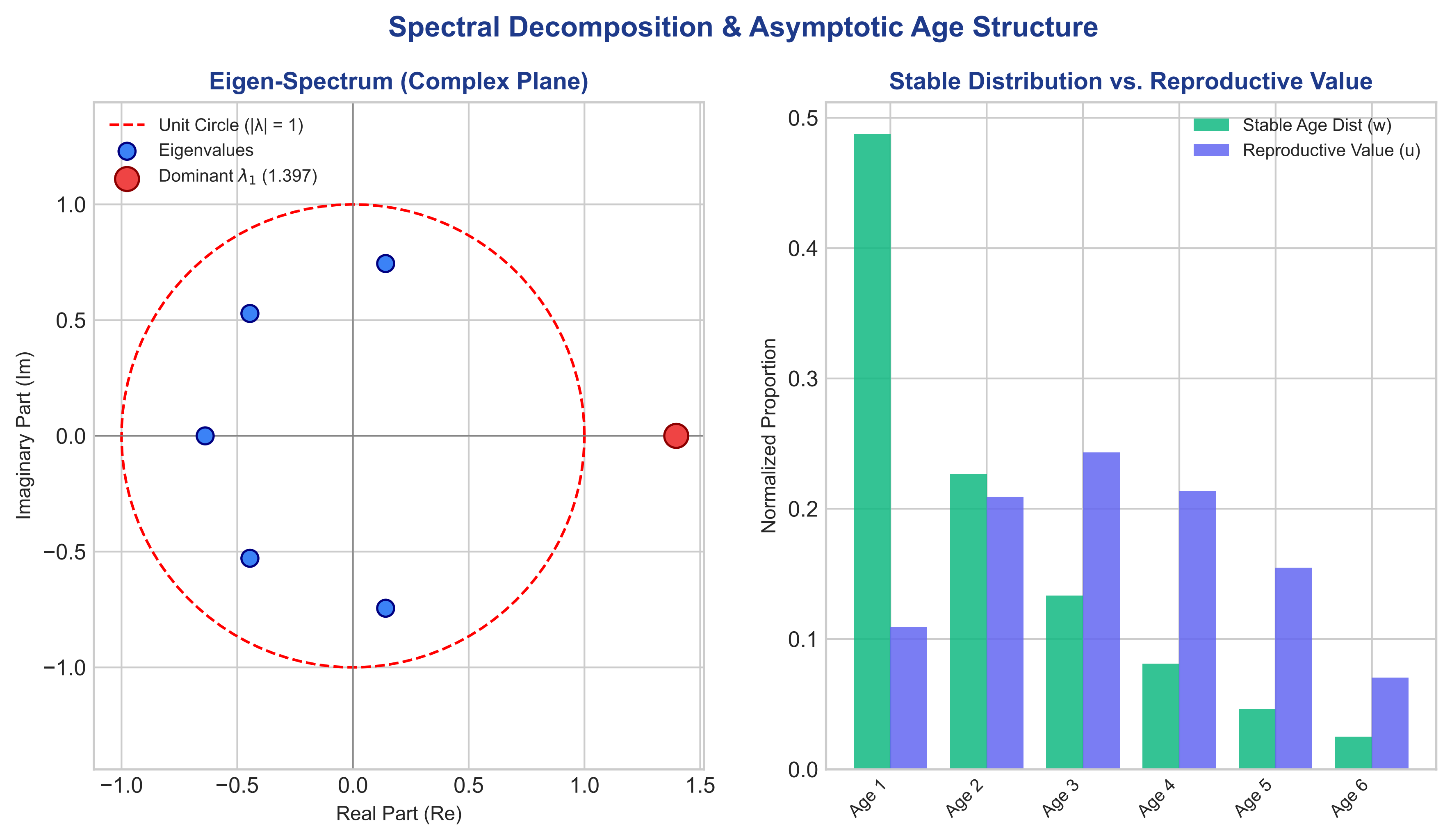
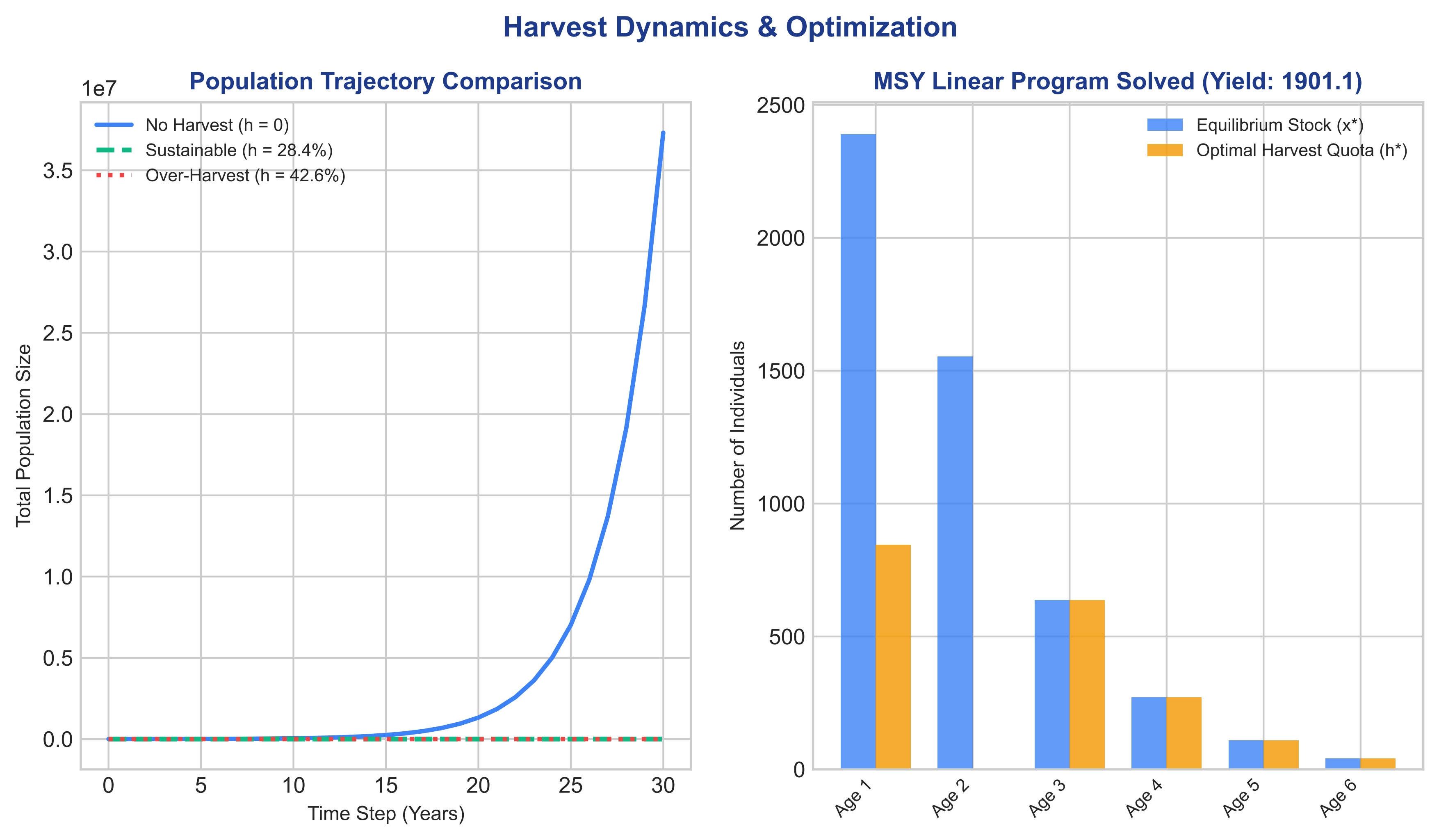
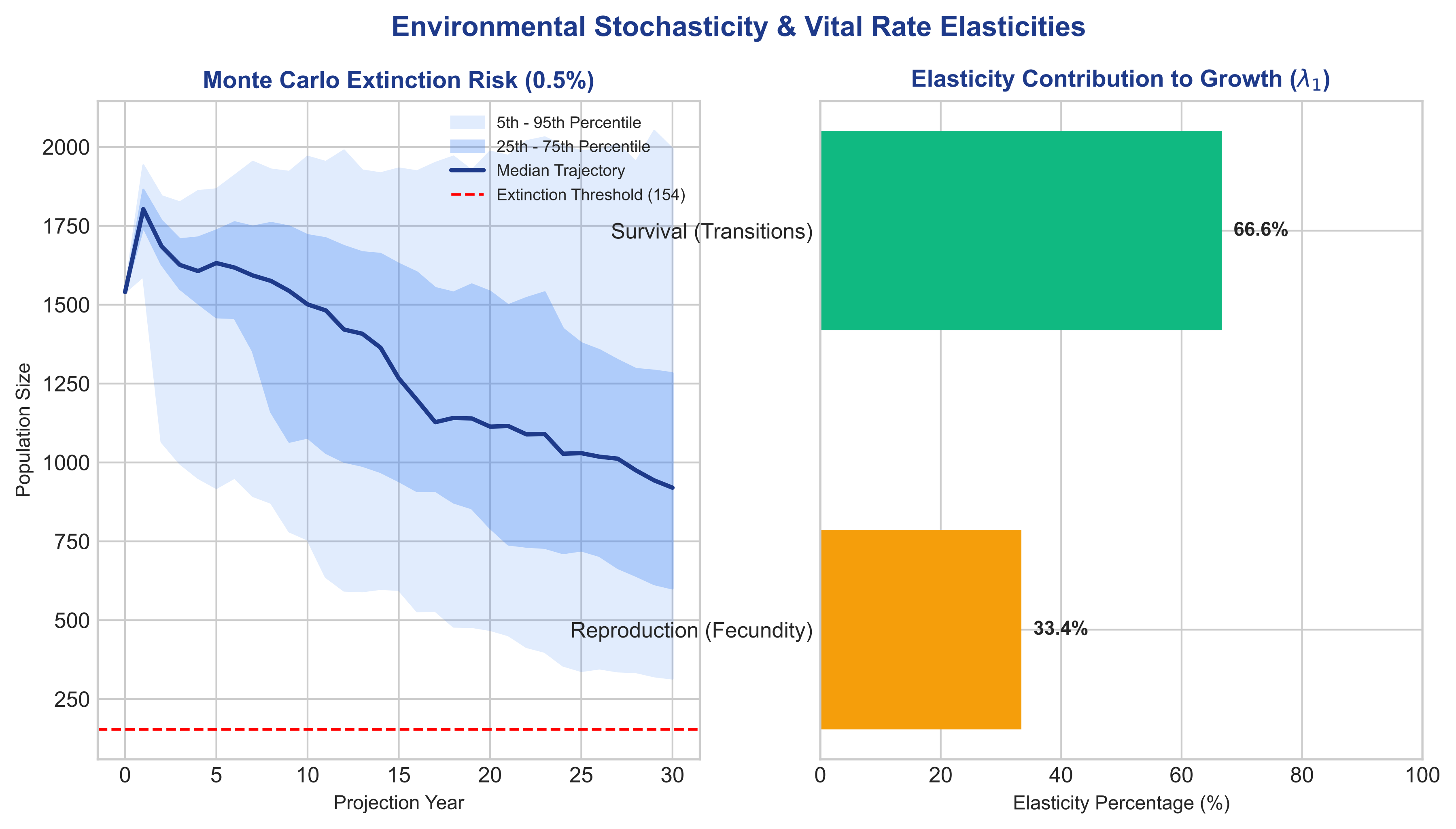

Spectral Decomposition & Asymptotic Age Structure
Eigen-Spectrum (Complex Plane)
Unit Circle (|λ| = 1)
Eigenvalues
Dominant λ_1 (1.397)
Imaginary Part (Im)
Real Part (Re)
1.0
0.5
0.0
−0.5
−1.0
1.5
Stable Distribution vs. Reproductive Value
Stable Age Dist (w)
Reproductive Value (u)
Normalized Proportion
0.5
0.4
0.3
0.2
0.1
0.0
Age 1
Age 2
Age 3
Age 4
Age 5
Age 6


Harvest Dynamics & Optimization
Population Trajectory Comparison
1e7
No Harvest (h = 0)
Sustainable (h = 28.4%)
Over-Harvest (h = 42.6%)
Total Population Size
Time Step (Years)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
0
5
10
15
20
25
30
MSY Linear Program Solved (Yield: 1901.1)
Equilibrium Stock (x*)
Optimal Harvest Quota (h*)
Number of Individuals
0
500
1000
1500
2000
2500
Age 1
Age 2
Age 3
Age 4
Age 5
Age 6


Environmental Stochasticity & Vital Rate Elasticities
Monte Carlo Extinction Risk (0.5%)
5th - 95th Percentile
25th - 75th Percentile
Median Trajectory
Extinction Threshold (154)
Population Size
Projection Year
0
5
10
15
20
25
30
250
500
750
1000
1250
1500
1750
2000
Elasticity Contribution to Growth (λ_1)
Survival (Transitions)
Reproduction (Fecundity)
66.6%
33.4%
Elasticity Percentage (%)
0
20
40
60
80
100